The weak acid is dissociated only to a small extent. Moreover, due to common ion effect, the dissociation is further suppressed and hence the equilibrium concentration of the acid is nearly equal to the initial concentration of the unionised acid. Similarly, the concentration of the conjugate base is nearly equal to the initial concentration of the added salt.

$$\left[H_3O^+\right]=K_a\frac{[acid]}{[salt]} \qquad ....(8.21)$$

Here [acid] and [salt] represent the initial concentration of the acid and salt, respectively used to prepare the buffer solution

Taking logarithm on both sides of the equation

$$\log [H_3O^+]=\log K_a+\log\frac{[acid]}{[salt]} \qquad ....(8.22)$$

reverse the sign on both sides

$$-\log [H_3O^+]=-\log K_a-\log\frac{[acid]}{[salt]} \qquad ....(8.23)$$

We know that

$pH = -\log [H_3O^+]$ and $pK_a = -\log K_a$

$$\Rightarrow pH = pK_a-\log\frac{[acid]}{[salt]} \qquad ....(8.24)$$

$$\Rightarrow pH = pK_a+\log\frac{[salt]}{[acid]} \qquad ....(8.25)$$

Similarly for a basic buffer, $pOH = pK_b+\log\frac{[salt]}{[base]}$ ....(8.26)

Example 8.6

1. Find the pH of a buffer solution containing 0.20 mole per litre sodium acetate and 0.18 mole per litre acetic acid. K_a for acetic acid is 1.8×10^{-5} .

$$pH = pK_a+\log\frac{[salt]}{[acid]}$$

Given that $K_a = 1.8 \times 10^{-5}$

$$\therefore pK_a = -\log(1.8\times10^{-5}) = 5-\log 1.8$$
$$= 5 - 0.26$$
$$= 4.74$$

$$\therefore pH = 4.74 + \log\frac{0.20}{0.18}$$
$$= 4.74 + \log {}^{10}\!/_{9} = 4.74 + \log 10 - \log 9$$
$$= 4.74 + 1 - 0.95 = 5.74 - 0.95$$
$$= 4.79$$

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